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## B.Tech. Degree II Semester Regular Examination July 2024

### EE/EC/CS/IT 23-200-0205B INTRODUCTION TO CYBER PHYSICAL SYSTEMS (2023 Scheme)

Time: 3 Hours

Maximum Marks: 50

## Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Understand the features and components of Cyber Physical Systems.

CO2: Understand the elementary constructs of Arduino Software.

CO3: Develop optimal programs and circuits for interfacing various sensors with Arduino.

CO4: Apply Arduino IDE for developing suitable programs for interfacing sensors and actuators with Arduino and Node MCU.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

**PART A**(Answer **ALL** questions)

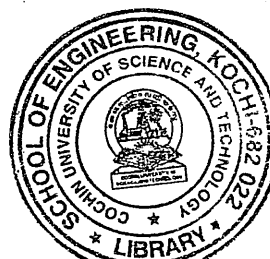
(5 × 2 = 10)

|                                                                               | Marks | BL | CO | PO |
|-------------------------------------------------------------------------------|-------|----|----|----|
| I. (a) Compare active and passive sensors in IoT using a suitable example.    | 2     | L4 | 1  | 1  |
| (b) Describe the general structure of an Arduino program.                     | 2     | L2 | 2  | 1  |
| (c) Contrast the analogRead() function with digitalRead() in Arduino.         | 2     | L4 | 2  | 1  |
| (d) Write a NodeMCU code to blink an LED at 1-second intervals.               | 2     | L6 | 4  | 3  |
| (e) How do you set up and use the Arduino IDE for programming Arduino boards? | 2     | L3 | 2  | 5  |

**PART B**

(4 × 10 = 40)

|                                                                                                                           |   |    |   |   |
|---------------------------------------------------------------------------------------------------------------------------|---|----|---|---|
| II. (a) Explain the features of Cyber Physical System.                                                                    | 6 | L2 | 1 | 1 |
| (b) What are the key characteristics that define the Internet of Things (IoT)?                                            | 4 | L2 | 1 | 1 |
| <b>OR</b>                                                                                                                 |   |    |   |   |
| III. (a) Define Cyber-Physical Systems (CPS). Explain the general structure of CPS.                                       | 6 | L2 | 1 | 1 |
| (b) List and describe some of the key enabling technologies that support IoT implementations.                             | 4 | L1 | 1 | 1 |
| IV. (a) Compare and contrast the applications of delay(), millis(), micros() and delayMicroseconds() in Arduino projects. | 4 | L4 | 2 | 1 |
| (b) Explain the various types of operators used in Arduino programming.                                                   | 6 | L2 | 2 | 3 |
| <b>OR</b>                                                                                                                 |   |    |   |   |
| V. (a) List and explain the basic data types supported by Arduino.                                                        | 5 | L1 | 2 | 1 |
| (b) Explain the functionality of these components on an Arduino board:                                                    | 5 | L2 | 2 | 3 |
| (i) Crystal oscillator.                                                                                                   |   |    |   |   |
| (ii) Voltage regulator.                                                                                                   |   |    |   |   |
| (iii) TX RX LEDs.                                                                                                         |   |    |   |   |
| (iv) AREF.                                                                                                                |   |    |   |   |



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|           |                                                                                                     | Marks | BL | CO | PO |
|-----------|-----------------------------------------------------------------------------------------------------|-------|----|----|----|
| VI.       | (a) Describe how PWM works on Arduino and its applications.                                         | 5     | L2 | 2  | 1  |
|           | (b) How do you integrate an ultrasonic sensor with Arduino for distance measurement?                | 5     | L3 | 3  | 3  |
| <b>OR</b> |                                                                                                     |       |    |    |    |
| VII.      | (a) Discuss the process of interfacing humidity and temperature sensors with Arduino.               | 6     | L3 | 3  | 3  |
|           | (b) What are pull-up and pull-down resistors and how do they affect push button switch interfacing? | 4     | L2 | 3  | 1  |
| VIII.     | (a) List the main differences between Arduino Uno and Node MCU development boards.                  | 5     | L1 | 4  | 1  |
|           | (b) Explain how to connect NodeMCU to a WiFi network.                                               | 5     | L2 | 4  | 5  |
| <b>OR</b> |                                                                                                     |       |    |    |    |
| IX.       | (a) Write an Arduino code to control a servo motor.                                                 | 5     | L6 | 3  | 3  |
|           | (b) How can you interface a gas sensor with NodeMCU?                                                | 5     | L3 | 4  | 5  |

Bloom's Taxonomy Levels

L1 – 14%, L2 – 43%, L3 – 19%, L4 – 14%, L6 – 10%:

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